12
QUESTIONS

~~Just ask nicely~~

Good prompts aren't magic. They're specs.

What interviewers really test.



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FOUNDATION 01

What Prompting Really Is

01

You don't retrain the model. You change **how you ask**. Interviewers want to hear that prompts are instructions with structure, not tricks.

Vague

"Explain APIs."

SPECIFIC

"Explain APIs to a beginner, simple words, one real-world example."

A STRONG PROMPT STATES

- The task, explicitly
- The output format
- Constraints: length, tone, source
- What to do if info is missing

> **Same model, better ask, far better answer.**



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FOUNDATION 02

Zero-Shot vs Few-Shot

02

Zero-shot gives the task with no examples. Few-shot adds a handful so the model copies the pattern.

Zero-shot

Fast and simple. Great for easy tasks like sentiment labels.

FEW-SHOT

Add 2 to 5 examples when format or accuracy matters.

When zero-shot wobbles

Inconsistent output means it is time for examples.

WHY INTERVIEWERS ASK

- Tests if you know prompt vs model
- Shows you can design assistants
- Reveals depth beyond definitions

Example

Classify: "I loved the movie" → Positive. Then show two labeled samples before the real input.

Start zero-shot. Add examples only when it fails.



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FOUNDATION 03

System vs Role Prompt

03

Candidates mix these up constantly. One sets the rules, the other sets the identity.

System prompt

Global rules: "Answer concisely, never speculate."

ROLE PROMPT

A persona: "You are a senior Python instructor."

1. **System: how the model behaves overall**
2. **Role: who the model acts as**
3. **Together: consistent voice plus consistent rules**

System = behavior. Role = identity.



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FOUNDATION 04

Temperature, Top-K, Top-P

All three shape which tokens the model may pick next.

Low (0 to 0.3)
Predictable, factual answers.

Medium (0.5 to 0.8)
Balanced creativity and accuracy.

High (1.0+)
Varied but riskier output.

Top-K
Fixed pool: only the K likeliest tokens.

TOP-P
Dynamic pool: tokens until probability hits P.

Top-K is a fixed cap. Top-P adapts.

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FOUNDATION 05

Fixing Loops and Hallucinations

Two classic failures, two sets of fixes.

MODEL REPEATS ITSELF

- Lower temperature
- Reduce top-K or top-P
- Set a max output length
- Say: "Give the answer once."

MODEL MAKES THINGS UP

- "Use only the context below"
- Allow "I don't know"
- Ask for justification from context

> **Constrain the model and give it an exit.**



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FOUNDATION 06

Chain of Thought

06

Ask the model to reason step by step before answering. It lifts accuracy on math, logic and multi-step problems.

Example

3 pens at $2 each: how much do I pay? Think step by step.

1. **Model lays out intermediate steps**
2. **Then commits to a final answer**
3. **Errors become visible and fixable**

Self-consistency

Sample several chains, keep the most common answer.

Show the work, get better answers.



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FOUNDATION 07

Tree of Thoughts

07

CoT walks a single path that may be wrong. ToT branches, scores, and backtracks.

Chain of Thought

One linear path. No way back from a dead end.

TREE OF THOUGHTS

Many branches. Weak ones are dropped early.

Example

"What general principle governs compound interest? Now compute 5 years at 8%."

Without it

Model jumps into details and gets stuck.

WITH IT

Model anchors on the rule, then applies it.

Planning task

Explore Plan A, B and C, then discard the weaker two.

Use ToT for planning and optimization, not simple Q&A.



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FOUNDATION 08

Step-Back Prompting

08

Before solving, ask the model to zoom out. It stops the model from diving into details too early.

1. **Step 1: What general principle applies?**
2. **Step 2: Now solve the specific question.**
3. **Check the answer against the principle**

CoT

Reasoning stays inside one flow.

REACT

Think, call a tool, observe, think again.

Best for

Complex or unfamiliar problems.

Principle first, details second.



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FOUNDATION 09

ReAct: Think, Act, Observe

CoT only reasons. ReAct reasons and **uses tools**: search, calculator, retriever.

1. **Thought: I need the latest leave policy**
2. **Action: search HR documents**
3. **Observation: updated policy retrieved**
4. **Thought: now I can answer**

Use it when

Answers need outside data or tools.

> **CoT thinks. ReAct thinks and acts.**



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FOUNDATION 10

Automatic Prompt Engineering

10

Let the system write and tune prompts instead of guessing by hand.

1. **Generate many prompt candidates**
2. **Test them on sample inputs**
3. **Compare outputs, pick the winner**

Example

"Extract Name, Email, Phone as JSON. Ignore all other text."

Judge by

Correctness, consistency, format, low hallucination.

EXAMPLE

Prompt A 90% correct beats Prompt B at 70%.

Measure prompts like you measure code.



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FOUNDATION 11

Structured Output and Variables

Production prompts need predictable shape and reusable templates.

FOR CLEAN JSON OR HTML

- Say "Return only valid JSON"
- Define keys or tags upfront
- Forbid extra explanations

template.txt

```
Answer using the context below.
Context: {context}
Question: {user_question}
```

Bias fix

Ask for balanced views. No assumptions about groups.

STYLE CONTROL

Name the tone and audience explicitly.

Variables let one prompt serve every user.



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FOUNDATION 12

Safety, Bias and Guardrails

12

Constraints and system rules are your seatbelts.

Constraints

Word limits, allowed sources, explicit refusal rules.

Neutrality

Ask for balanced, inclusive, factual language.

System-level rules

Override risky user input in sensitive tasks.

Example

"You are a compliance assistant. No legal or medical advice. Refer users to a professional."

Safe behavior is designed, not hoped for.



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INTERVIEW READY

Prompting is engineering, not luck.

Explain the why behind every technique and you will stand out.

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